

Cloud Computing Tutorial - 4 and 5

CloudSim Simulation

Observation:

Tutorial 4 & 5

Q1) what is cloudsim?
- cloudsim is a simulation tool that helps research and students study how cloud computing env. work without needing real cloud infra.

Q2) what does cloudsim do?
- cloudsim allows users to:
① simulate cloud infrastructure.
② Testing scheduling & resource allocation.
③ Evaluating performance.
④ optimize cloud resources.

Prerequisites - JDK 8 or higher.
- Guice.
- cloudsim source code.
- Apache commons math lib.

Run a cloudsim Example:-
eg:- cloudsim Example1.java in source folder output will display in console.

Output:-

Starting cloudsim Example1.
Initializing.
Starting cloudsim version 3.0.
Datacenter-0 is starting.
:
Simulation completed.
Output status -> success.
cloudsim Example1 finished.

CloudSim Simulation:

```
C:\Users\Admin>java --version
java 23.0.1 2024-10-15
Java(TM) SE Runtime Environment (build 23.0.1+11-39)
Java HotSpot(TM) 64-Bit Server VM (build 23.0.1+11-39, mixed mode, sharing)

C:\Users\Admin>
```

Fig 1.1: JDK installation verification

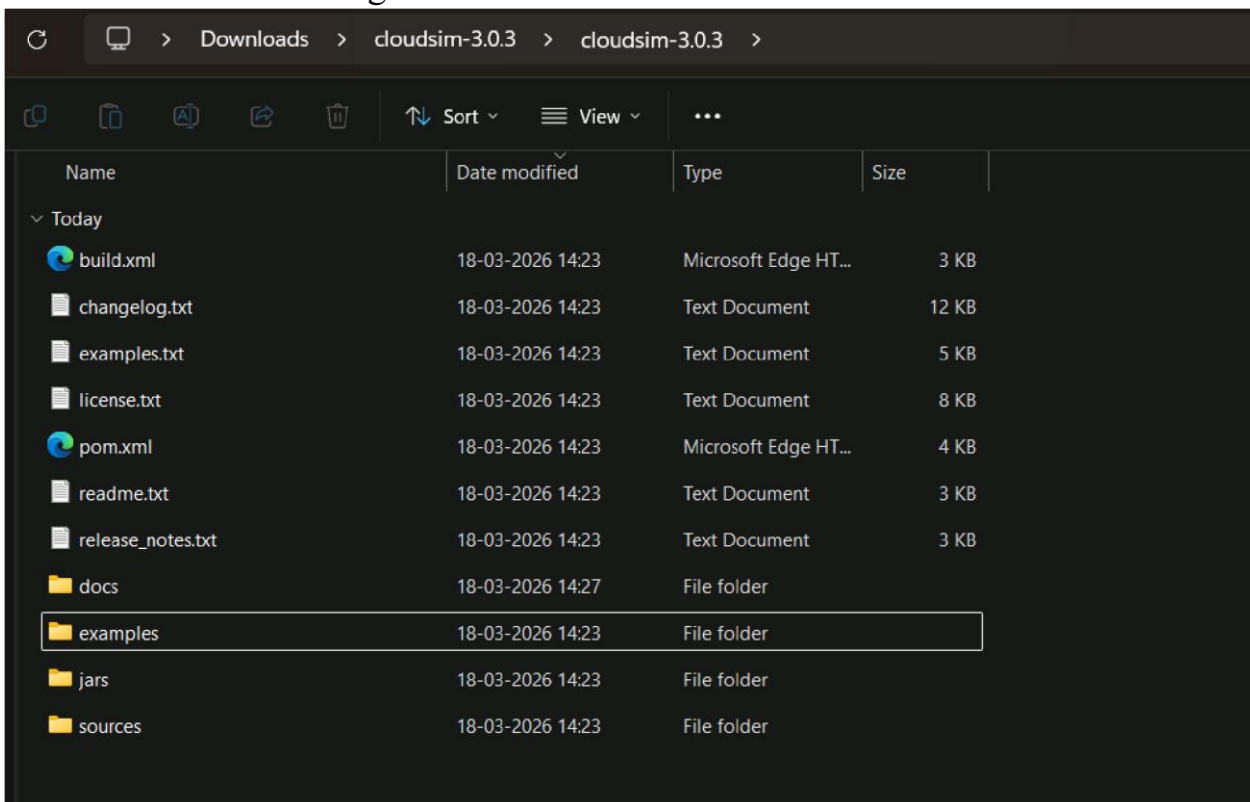


Fig 1.2: cloudsim zip folder extraction

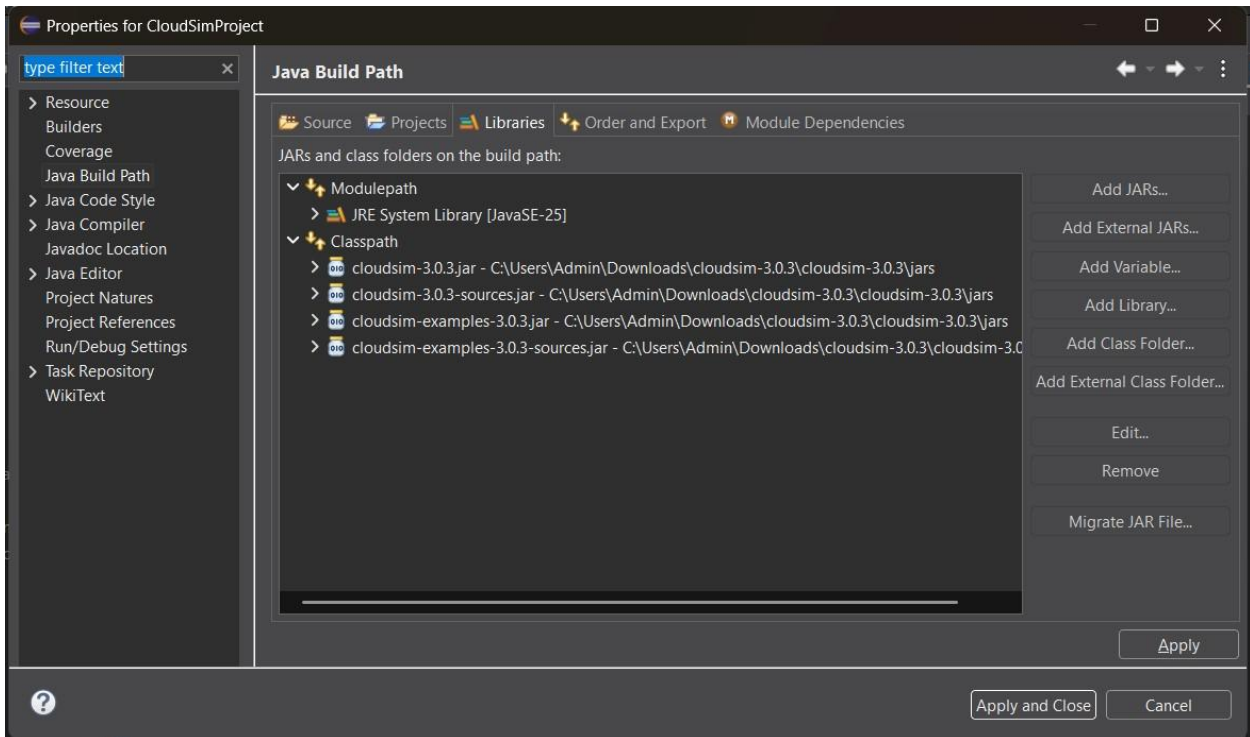


Fig 1.3: CloudSimProject properties

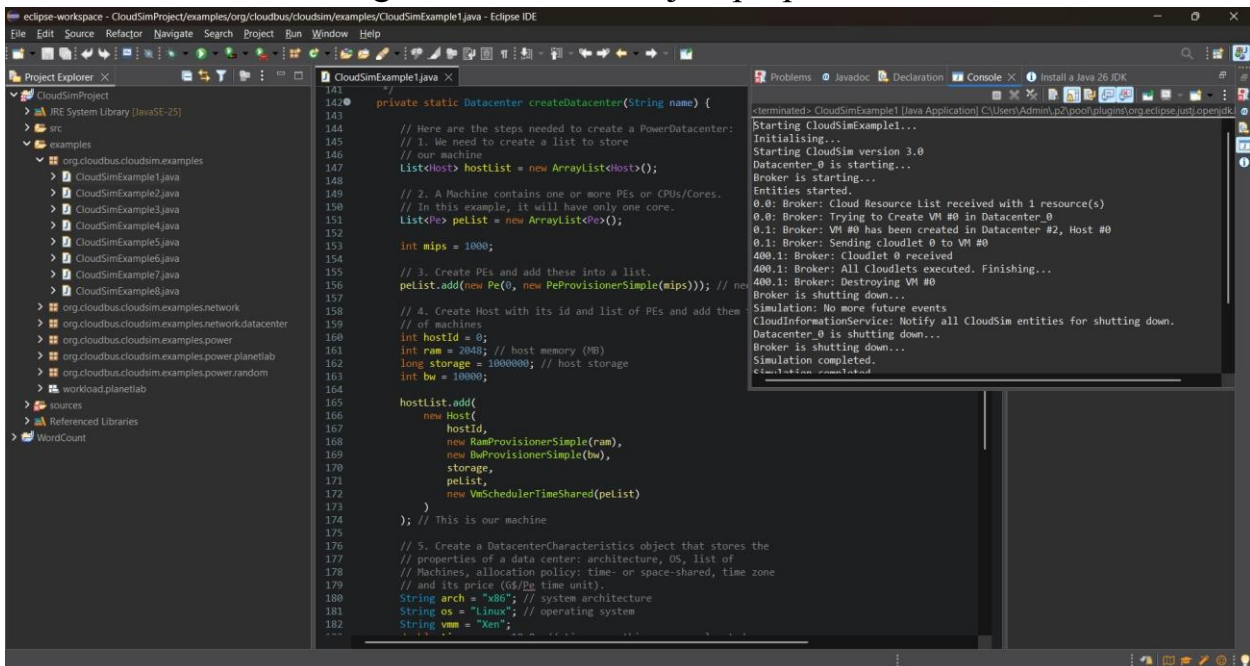


Fig 1.4: Running CloudSimProject - CloudSimExample1.java program

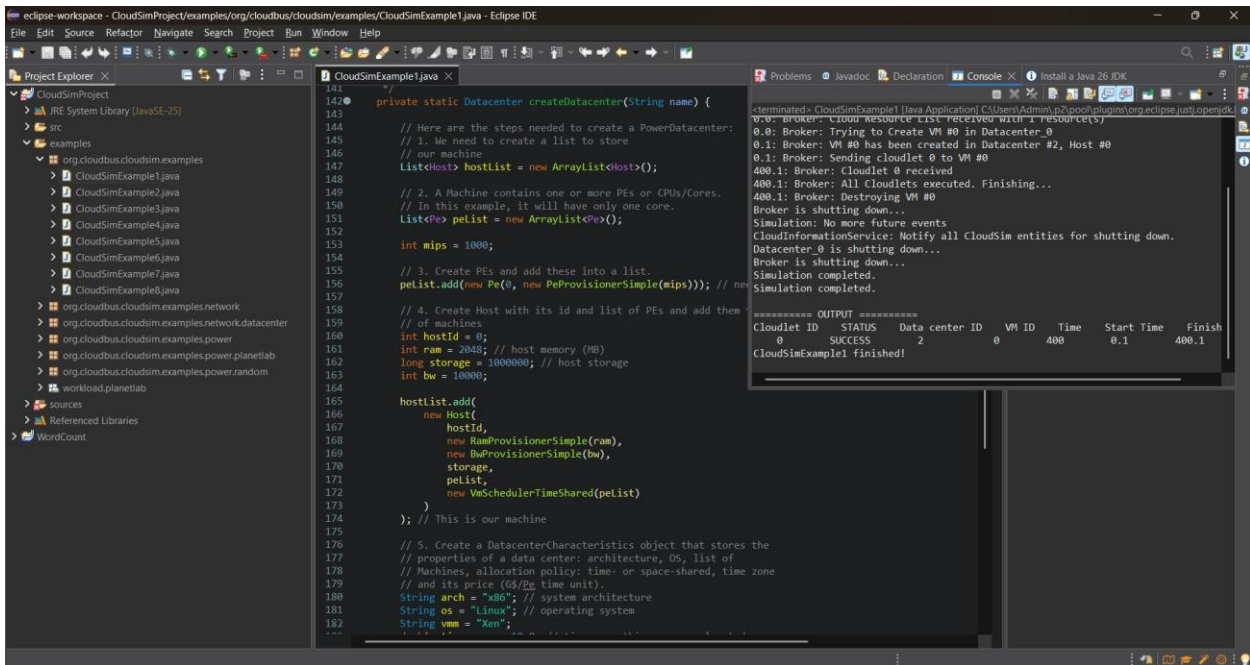


Fig 1.5: Simulation successful